Customer Segmentation Using Cluster Analysis

Done by

Mahima Srinivasan (101546056)

Velon Muruguthas (101562891)

Ashwin Balaji (101539897)

Purusothaman Loganathan (101553143)

Lalith Narayanan Janarthanan (101571149)

Tejaswi Rongali (101571146)

Introduction

In today's data-driven retail environment, understanding customer behavior is crucial for devising effective marketing strategies. Customer segmentation enables businesses to categorize their clientele based on various characteristics and target them more effectively. This report explores a cluster analysis approach to segment customers using data related to their annual income and spending scores. The objective is to derive meaningful patterns and suggest actionable strategies for a retail mall.

Business Objective

To categorize customers into distinct groups using clustering methods to allow for more personalized marketing efforts, thus enhancing customer satisfaction and increasing profitability.

Hypothesis

- Customers with higher annual income tend to have higher spending scores.
- There exists a group of low-income, high-spending customers.
- Middle-income customers display diverse spending patterns.

Methodology

To effectively segment customers, the following steps were undertaken:

Tools Used:

- Python programming language
- Libraries: Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn

Data Preprocessing:

- Missing values and outliers were checked and handled.
- Selected features: Annual Income and Spending Score
- Data was standardized using StandardScaler to ensure uniformity.

Clustering Technique:

- **K-Means Clustering** was chosen due to its efficiency with numerical data and scalability.

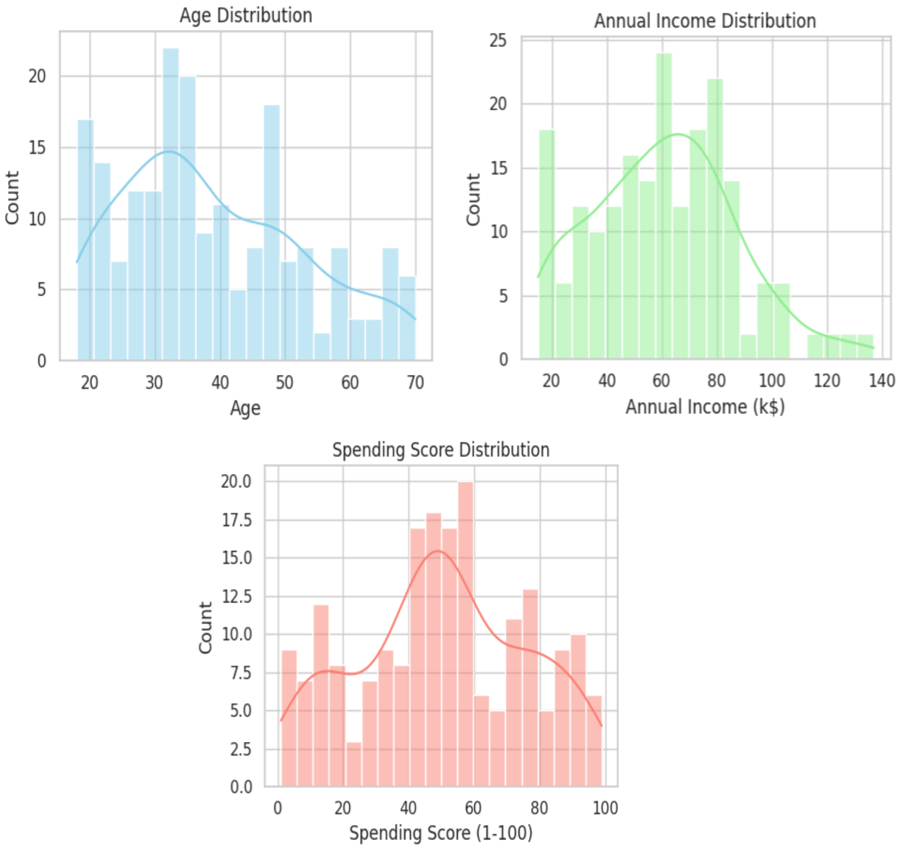
Steps Followed:

1. Data exploration through visualizations.
2. Use of the **Elbow Method** to determine the optimal number of clusters.
3. Application of the K-Means algorithm with the selected number of clusters.
4. Visualization and interpretation of the clusters.

Exploratory Data Analysis (EDA)

Initial visualizations helped understand customer distribution:

- **Distribution of Age, Annual Income, and Spending Score**



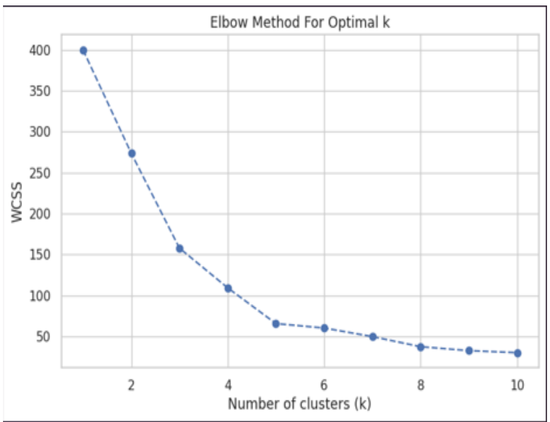
Age distribution reveals a high concentration between 20 to 40 years.

- Annual income is right-skewed, with most values clustered around 60k$.
- Spending score is uniform but peaks near 50, indicating average spenders dominate the dataset.

Determining Optimal Clusters

To find the optimal number of clusters, we employed the **Elbow Method**, which examines the Within-Cluster Sum of Squares (WCSS).

Elbow Method for Optimal K



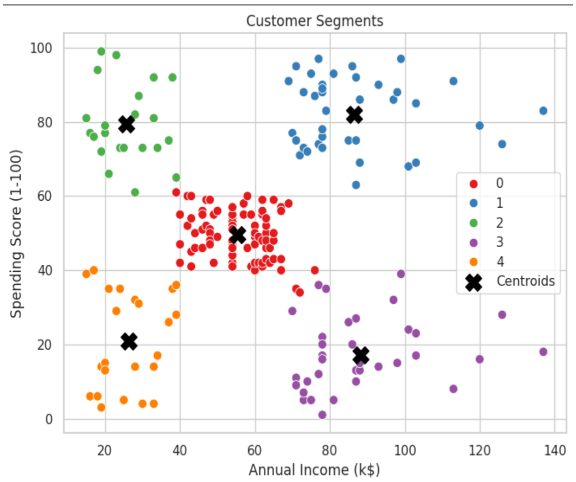
The curve bends at **k=5**, indicating this as the optimal number of clusters.

- Beyond k=5, WCSS declines more gradually, offering minimal improvement in cluster compactness.

Cluster Analysis and Results

With k=5 determined, we proceeded to apply the K-Means algorithm.

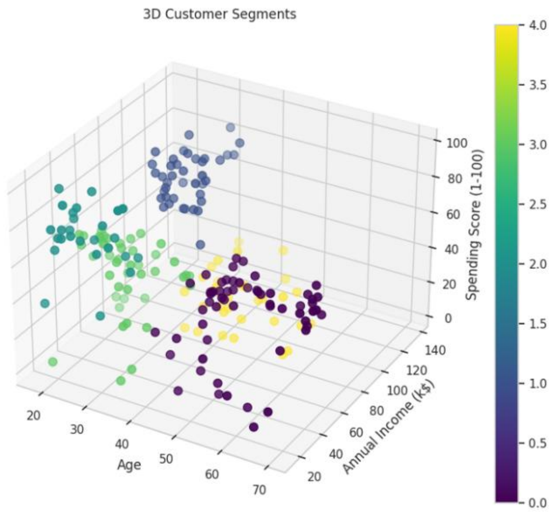
- **3D Customer Segments**



- By incorporating Age as a third axis, this plot provides enhanced visual separation between clusters.
- Useful in identifying age-related spending trends.

- Displays five distinct clusters based on Annual Income and Spending Score.
- Centroids represent the average customer in each segment.

- **Cluster Labels with Insights**



Cluster	**Label**	**Interpretation**
1	High Income, Low Spenders	Value-oriented customers with savings habits
2	Low Income, High Spenders	Impulsive or brand-loyal, budget-insensitive
3	Low Income, Low Spenders	Cost-conscious or financially constrained
4	High Income, High Spenders	Ideal segment for premium offerings
5	Middle Income, Middle Spenders	Stable, reliable customer group

Hypothesis Evaluation

Hypothesis	Outcome
High income → high spending	❌ Not universally true. Two clusters show split behavior.
Low-income → high spending group exists	✅ Confirmed. One cluster exhibits this pattern.
Middle income → most diverse spending	✅ Confirmed. Shows both low and moderate spending levels.

The analysis supports the idea that spending behavior is influenced by multiple factors beyond income, including personal preferences, lifestyle, and psychological traits.

Business Implications

With the customer base segmented, the retail mall can implement more targeted strategies:

- Cluster 1 (High Income, Low Spenders): Encourage engagement through VIP perks or lifestyle-related campaigns.
- Cluster 2 (Low Income, High Spenders): Budget-conscious promotions, bundling, loyalty discounts.
- Cluster 3 (Low Income, Low Spenders): Focus on price sensitivity, clearance sales.
- Cluster 4 (High Income, High Spenders): Premium loyalty programs, exclusive deals, early product launches.
- Cluster 5 (Middle Income, Middle Spenders): Regular promotions and steady engagement through seasonal offers.

Limitations and Future Work

- Only two features were used (Annual Income and Spending Score), excluding demographic or behavioral variables.
- K-Means assumes spherical clusters and equal variance, which may not reflect real-world complexity.

Future Recommendations:

- Include features like Age, Gender, Frequency of Visits.
- Consider more advanced algorithms (e.g., DBSCAN, Hierarchical Clustering).
- Periodically re-run clustering to adapt to evolving customer behavior.

Conclusion

Customer segmentation using K-Means has proven effective in identifying meaningful customer groups. These insights empower the business to tailor marketing strategies, optimize promotions, and enhance customer experience. Continuous refinement and integration with broader datasets will further elevate the effectiveness of segmentation strategies.